Rodrigo Rojas 98.6%

1) a)
$U_f = 298.49$ btu/lb
$U_g = 1105.1$ btu/lb
$h_f = 298.51$ btu/lb
$h_g = 1187.5$ btu/lb
$v_f = 0.01774$ ft³/lb
$v_g = 4.4327.$ ft³/lb
$U_1 = 500$ btu
$T_1 = 327.81$ °F

$U_1 = U_f + x(U_g - U_g)$
$500 = 298.49 + x_1(1105.5 - 298.49)$
$x_1 = 0.25$
$h_1 = h_f + x(h_f - h_g)$
$= 298.51 + 0.25(1187.5 - 298.51)$
$\boxed{h_1 = 520.75 \text{ btu/lb}}$
$v_1 = v_f + x_1(v_g - v_g)$
$= 0.01774 + 0.25(4.4327 - 0.01774)$
$\boxed{v_1 = 1.121 \text{ ft}^3/\text{lb}}$

B)
$U_f = 85.98$ btu/lb
$U_g = 564.42$ btu/lb
$v_f = 0.02553$ ft³/lb
$v_g = 3.9664.$ ft³/lb
$x = 0.5$
$\boxed{P_2 = 73.354}$

$v_2 = v_f + x_2(v_f - v_g)$
$= 0.02537 + 0.5(3.9664 - 0.02553)$
$\boxed{v_2 = 2 \text{ ft}^3/\text{lb}}$
$U_2 = 85.98 + 0.5(564.42 - 85.98)$
$\boxed{U_2 = 327.2 \text{ btu/lb}}$

c)
$T_3 = 480$ °C
$U_3 = 3025.7$ kJ/kg
$h_3 = 3348.4$ kJ/kg

d)
$P_4 = 1.7299$ bar
$v_4 = 0.1464$ m³/kg
$U_4 = 2148.84$ kJ/kg

2)
$T_1 = 180$
$\boxed{P_1 = 1 \text{ MPA}}$
$T_2 = 100$
$P_2 =$
$P_2 = 1 + 100 - 99.63 \frac{1.5-1}{111.4-99.63}$
$P_2 = 1.0157$ bar

$v_2 = v_1 = 0.1941$ m³/kg
$v_f = 0.0010437$ m³/kg
$v_g = 1.6718$ m³/kg
$v_2 = v_f + x_2(v_g - v_f)$
$0.1941 = 0.00104 + x(1.6718 - 0.00104)$
$\boxed{x = 0.116}$

3) $m = 2kg$ $\Delta P = 0$ $P_1 = P_2 = 300\ kPa$

$V_2 = 2.064$

a) $t_1 = ?$

table T H_2O @ 3 bar = 133.6 C

$T_1 = 133.6\ °C$

b) T_2

$v_2 = \frac{2.064}{2\ kg}$

$v_2 = 1.032$

$\boxed{T_2 = 400C}$

c) W $W = P\Delta V$

$= 300kPa\,(2.064 - 2(0.6058))$

$W = 2{,}557\ J$

4) $P_1 = 10\ bar$ $P_2 = 20\ bar$

$T_1 = 60\ C$ $T_2 = 30\ °C$

$m = 1\ kg$ $v_2 = 0.01187\ \frac{m^3}{kg}$

$W_{12} = 20\ kJ$ $u_2 = 225.14\ \frac{kJ}{kg}$

$v_1 = 0.02301$ $Q = W_{12} + m(u_2 - u_1)$

$u_1 = 270.32$ $Q = 20 + 1(255.14 - 270.32)$

$\boxed{Q = 4.82\ kJ}$

$V_1 = mv_1 = 1(0.02301)$

$\boxed{V_1 = 0.02301\ m^3}$

5) $m = 1.5$

$T_1 = 70\,C$ $\quad$ $T_2 = 120\,C$

$\boxed{P_1 = 31.19\ kPA}$ $\quad$ $v_2 = 0.89133\ m^3/kg$

$u_2 = 2528.9\ kJ/kg$

$v_1 = v_2 = 0.089133\ m^3/kg$

$v_{f1} = 0.0010228\ m^3/kg$

$v_{g1} = 5.042\ m^3/kg$

$0.89133 = 0.0010228 + x_1(5.042 - 0.0010228)$

$x = 17.66\%$

$u_1 = 292.95 + 0.1766(2469.6 - 292.95)$

$u_1 = 677.35\ kJ/kg$

$Q_{12} = W_{12} + \Delta U$ $\quad$ $W_{12} = 0$

$= 2528.9 - 677.35$

$q = 1851.55\ kJ/kg$

$Q = 1851.55(1.5)$

$\boxed{Q = 2777.325}$

6) $v_f = 0.8352E^{-3}\ m^3/kg$ $\quad$ $v_g = 0.0236$ $\quad$ 76

$v = 0.8352E\text{-}3 + 0.6(0.0236 - .8352E\text{-}3)$

$v_2 \quad 0.01444\ m^3/kg$

$0.6 = \frac{m_g}{m_g + 25}$

$m_g = 37.5\ kg$

$V_g = 0.0236 \times 37.5 = 0.885\ m^3$

$V_L = 25(0.8352E\text{-}3) = 0.02088\ m^3$

$V = (25 + 37.5)\ 0.01444 = \boxed{0.9056\ m^3}$

$X_v = \frac{0.885}{0.9056} = \boxed{0.97725}$

7) $m_{HE} = 0.2\ g$

$P_1 = 1.3\ bar$ $T_1 = 300K$

$P_2 = 0.8\ bar$ $T_2 = 290K$

$R_{HE} = \frac{8.314}{4} = \boxed{2.0785\ \frac{kJ}{kg}K}$

$PV = mRT$

$V_1 = \frac{0.2E\text{-}3 \cdot 2.0785 \cdot 300}{1.3E2}$ $V_2 = \frac{0.2E\text{-}3\ (2.0785)\ 290}{0.8E2}$

$\boxed{V_1 = 959\ cm^3}$ $\boxed{V_2 = 1506\ cm^3}$

8) $T_1 = 298\ K$

$P_1 = 275\ kPA$

$V_1 = 0.05\ m^3$

$T_2 = 530°K$

$Q_{12} = -2.75\ kJ$

$R = 0.1889\ \frac{kJ}{kg\ K}$

$275(0.05) = m(0.1889)(298)$

$\boxed{m = 0.2442\ kg}$ $C_v = 0.657\ \frac{kJ}{kg\ K}$

$-2.75 = W_{12} + 0.2442(0.657)(530 - 298)$

$\boxed{W = -39.4719\ kJ}$

X 808